k = 4 index calculus on $E(\mathbb{F}_{p^4})$, measured

relations $\propto p^{0.96}$, Wiedemann $\propto p^{2.04}$, rho $\propto p^{1.93}$; crossover from the fits 2^{98} ($h = 4$), sensitivity 2^{82}–2^{127}

`research/gaudry_quartic_d4_exponent_20261015` · 2026-10-15 · class: measurement

Summary. Gaudry/Diem index calculus with FGHR's D_4 oracle (PR #1863) and rho with the negation map were run end to end in ecbench on `fp4_torsion` curves for 12 primes $p = 29 \ldots 1009$: all 72 planned jobs, 312 records, all verified ($[d]G = Q$), `S` a count. Fitted over the rungs, the relation phase grows like $p^{0.96}$ (95 %: 0.65–1.27), the linear algebra (Wiedemann, new here) like $p^{2.04}$ (1.85–2.22), and rho like $p^{1.93}$ (1.78–2.09), as the analysis says (q^1, q^2, q^2). Index calculus still loses to rho at every measured size ($9.1 \cdot 10^5\times$ at $p = 29$, $6.7 \cdot 10^4\times$ at $p = 1009$). From the measured fits, it meets rho at a group of $2^{97.9}$ ($h = 4$; bootstrap $2^{94.6}$–$2^{102.5}$); with the relations slope at its interval ends, $2^{82.1}$–$2^{127.4}$. The §11.16 model gave $2^{105.4}$–$2^{106.5}$. **This extrapolates 14.5 bits in p, 58 bits in group size, beyond the top rung.** At that point the relation phase is still 97 % of the cost. Double large primes lower the fitted linear-algebra slope to 1.70 (interval 1.25–2.15, consistent with both the theory's 1.5 and the plain 2.04) but cost 1.09–2.31× more on paired walks at $p \geq 61$, and move the fitted crossover later (2^{128}). The first 60 jobs gave $2^{100.4}$; the last 12 jobs were run afterwards at the owner's request (§Last 12 jobs).

1. Method

Protocol `PROTOCOL.md`, registered at `08722a2b7` before any session; amendments: (1) queue reordered and stop extended to 16 h (runtime, before any IC record); (2) audit scope by size; (3) relaunch of 20 jobs after the container's CPU set shrank; (4) the 12 jobs that missed the stop run at the owner's request, with a corrected count. Code `0bea75038`. Arms (all `fghr-y-line:zero=1`, oracle `gaudry-d4:macaulay=26,weighted=1,chains=1,sparsest=1,replay=1,charge_unsolved=1`): `ic-d4-w` (`linalg=wiedemann`), `ic-d4-lp2`/`ic-d4-lp1` (`large_primes=2`/`1`, small base the first $\lceil N^\theta \rceil$ columns, $\theta = \frac{3}{4}, \frac{7}{9}$), `rho-neg`. One spec per (arm, p, target); the IC arms on a target share the walk. 60 of 72 jobs ran before the 16 h stop; the other 12 (third plain targets at five small rungs, second LP2 targets at seven) were run afterwards at the owner's request (amendment 4, same specs and binary). Earlier versions of this report said 59 and 13: that was a miscount.

New code. `linalg=wiedemann` solves for the target through a left-kernel vector of the relation matrix, $O(m \cdot \mathrm{nnz} + m^2)$ multiplications mod r, priced as `row_ops`; the default dense elimination ran on the same rows as an unpriced shadow and **gave the same logarithm in all 60 IC runs**. The large-prime wrapper works on any summand oracle; the existing large-prime code is binary-field only.

Fits. OLS of $\ln(\text{rung mean})$ on $\ln p$; interval = union of the Student-t 95 % interval and a bootstrap (10,000, targets resampled within rungs). The crossover solves $\sum_k A_k p^{\beta_k} = \overline{S}_{\text{rho}} \sqrt{p^4/h}$.

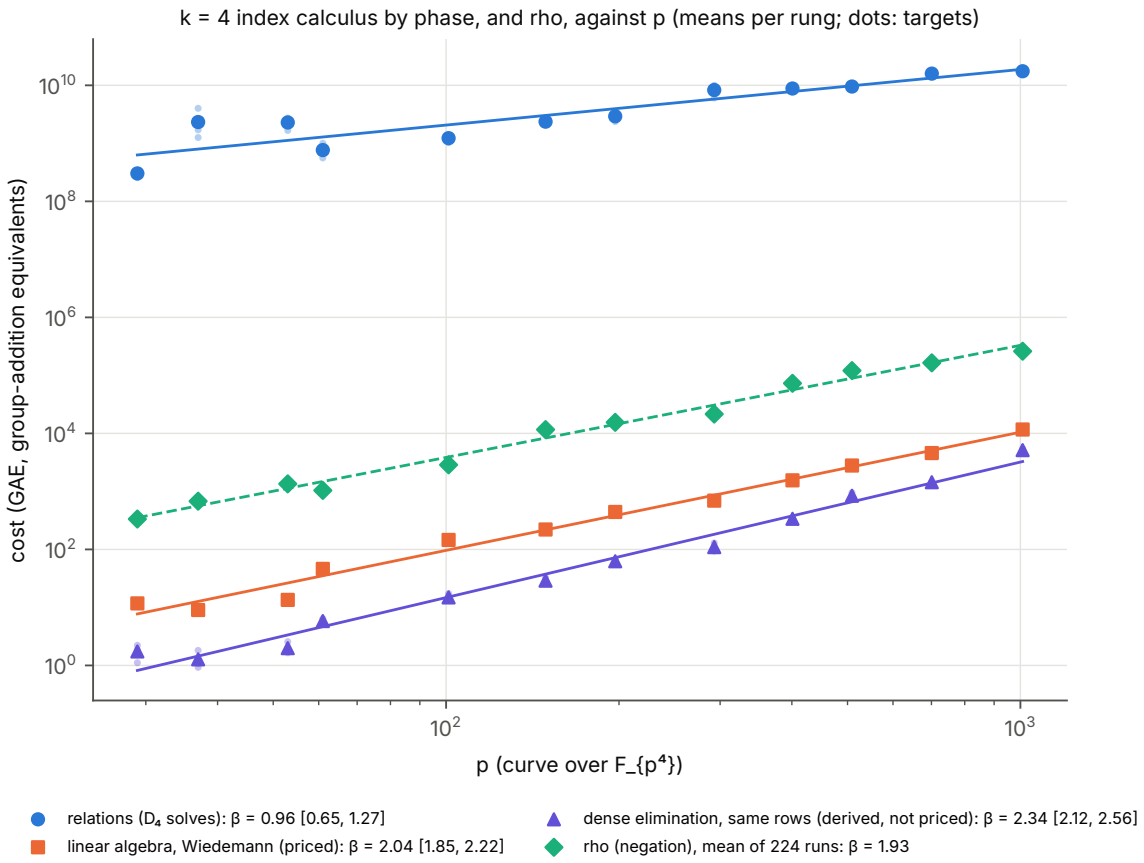


Figure 1: Cost by phase and rho against p (rung means; light dots: single targets). The dense elimination is derived from its `row_ops` at the same price and is not part of `S`.

2. Results

quantity	β	**95 % interval**	**expected**
relations phase (= total)	**0.96**	0.65–1.27 (bootstrap 0.90–1.03)	q^1
trials	0.85	0.54–1.17	
linear algebra, Wiedemann	**2.04**	1.85–2.22	q^2
dense elimination, same rows (derived)	2.34	2.12–2.56	q^3 worst case
rho (224 distinct runs)	**1.93**	1.78–2.09	q^2

The relations interval is wide because trials per relation vary between curves (rung mean trials over mean relations: 56 at $p = 29$, 75 at 61, 383 at 53, 496 at 37, 81 at 1009), which the within-rung bootstrap does not see. One D_4 solve costs $6.1 \cdot 10^5$ GAE at $p = 29$ and $8.3 \cdot 10^5$ at $p = 1009$; 171 of 220,652 systems (0.08 %) were degenerate and are charged. $\overline{S}_{\mathrm{rho}} = 1.08$ (1.02–1.15), drifting from 1.6 at $p = 29$ to 0.9–1.0 at the top.

crossover (group size)	$h = 2$	$h = 4$	$h = 16$
measured fits, point	$2^{96.0}$	$\mathbf{2^{97.9}}$ ($p^* = 2^{24.5}$)	$2^{102.0}$
bootstrap within rungs	$2^{92.8}$–$2^{100.4}$	$2^{94.6}$–$2^{102.5}$	$2^{98.4}$–$2^{106.8}$
relations slope at 0.65 / 1.27	$2^{80.6}$–$2^{124.6}$	$\mathbf{2^{82.1}}$–$\mathbf{2^{127.4}}$	$2^{85.2}$–$2^{133.2}$
§11.16 model (#1863)	$2^{103.0}$–$2^{103.7}$	$2^{105.4}$–$2^{106.5}$	—

With rho's own fitted power law in p: $2^{105.4}$ ($2^{99.9}$–$2^{113.7}$). The top rung is a $2^{39.9}$ group, so the crossover is **58 bits of group size (14.5 bits of p) beyond the measurements**, 7.5–8.6 bits of group size earlier than the model. The honest interval contains the model's figure; the bootstrap interval does not see between-curve scatter and understates the uncertainty. With the dense elimination priced instead, the cost meets rho at $2^{98.9}$ ($h = 4$) and rises above it again at 2^{128}; at $h = 16$ it meets it at $2^{106.3}$ and is above again from 2^{114}. With Wiedemann the fitted cost stays below rho through the extrapolated range ($p \le 2^{34}$).

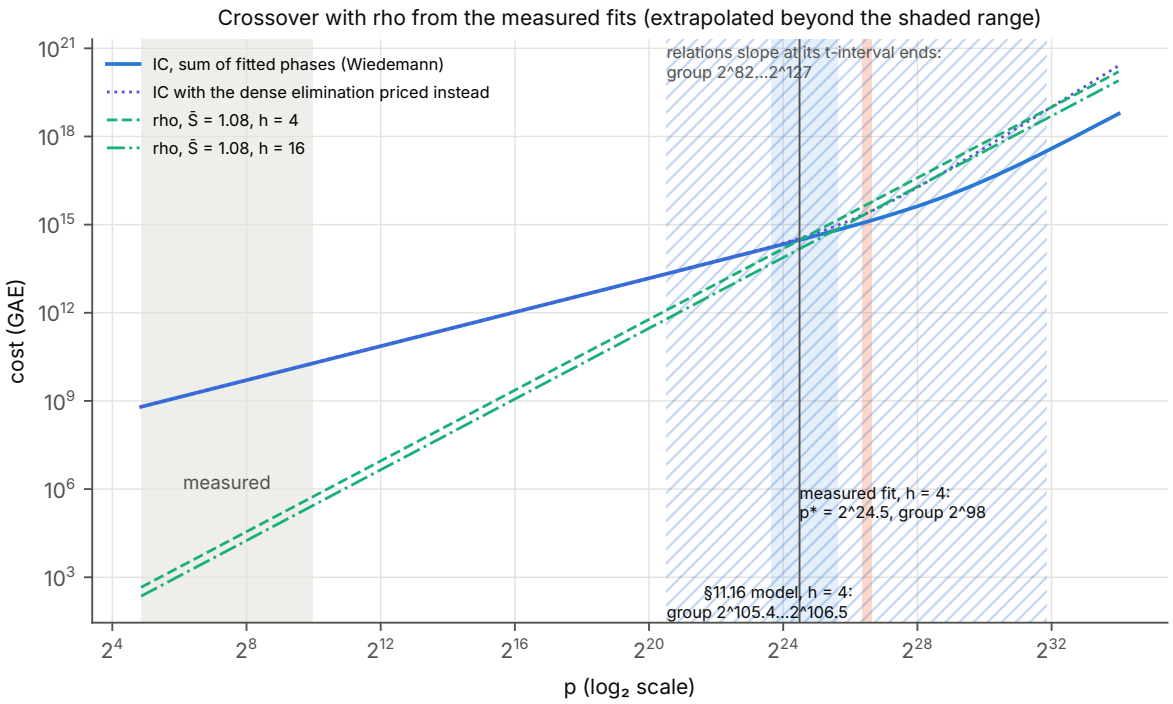


Figure 2: Extrapolation of the fitted phases to the crossover with rho. Grey: the measured range. Blue band: bootstrap; hatched: relations slope at its interval ends; orange: the §11.16 model at $h = 4$.

3. Large primes

arm	runs (rungs)	lin. alg. β	relations β	crossover $h = 4$
none	32 (12)	2.04 (1.85–2.22)	0.96 (0.65–1.27)	$2^{97.9}$
two, $\theta = \frac{3}{4}$	16 (8, $p \le 293$)	**1.70** (1.25–2.15)	1.26 (0.65–1.86)	$2^{127.7}$
one, $\theta = \frac{7}{9}$	12 (6, $p \le 149$)	1.62 (1.11–2.14)	1.49 (0.43–2.55)	2^{178} (153–214)

Paired S ratios (same target and walk), two large primes, two targets per rung: 1.00/1.00, 1.00/1.00, 0.91/1.12, 1.09/1.36, 1.09/1.32, 1.22/1.80, 1.80/2.31, 1.44/2.28 for $p = 29 \ldots 293$; one: 0.89–1.12 at $p \le 53$, 1.78–3.45 at $p \ge 61$. The matrix shrinks as the analysis says (Wiedemann 3.1–4.7× cheaper at $p = 293$) and the fitted slope falls from 2.04 to 1.70, but the intervals (1.25–2.15, 1.85–2.22) overlap, so it is consistent with the theory's 1.5 and with 2.04. The matrix is 10^{-6} of the cost here and 3 % at the crossover; the extra trials land on the relations phase. Large primes would pay only where the linear algebra dominates, past about 2^{117}. With the relations slope at 1.86 the double-large-prime cost never meets rho below $p = 2^{64}$.

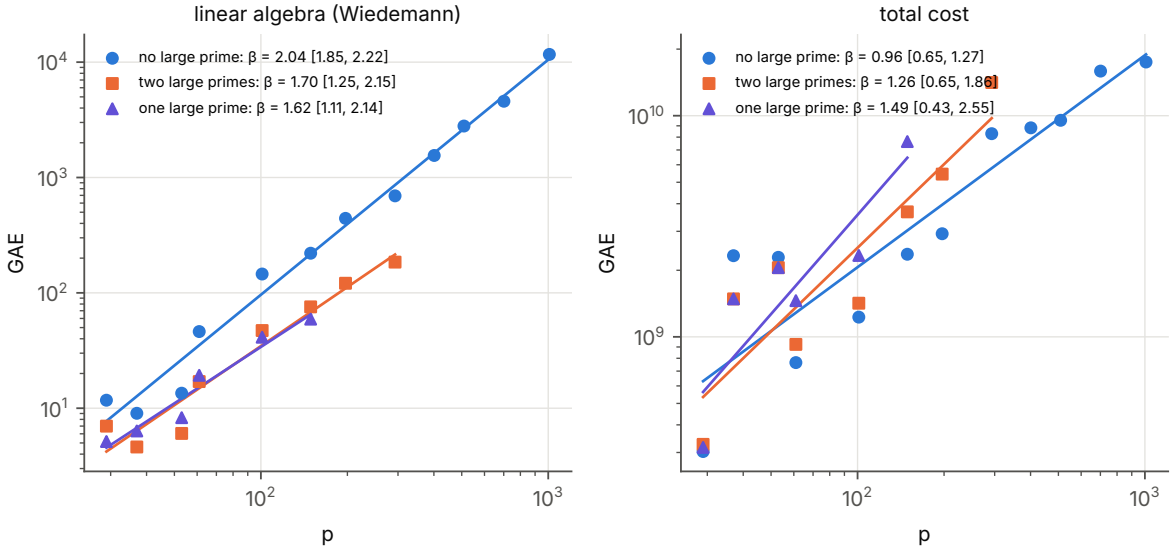


Figure 3: Linear algebra and total cost with and without large primes.

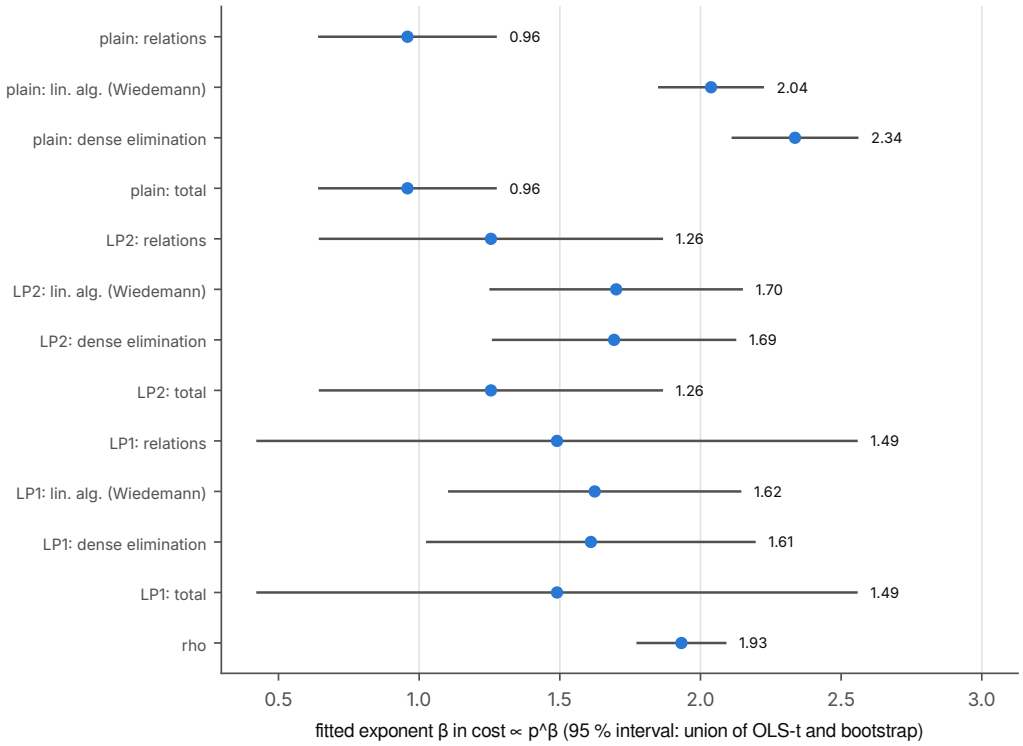


Figure 4: All fitted exponents with their 95 % intervals.

4. Last 12 jobs: the fit before and after

The 60-job fit (`raw/summary-60-jobs.json`, commit `9fb5c60e6`) was computed and read before the decision to run the remaining 12; it is kept. The 72-job fit is the headline because it uses the full planned set.

quantity	**60 jobs**	**72 jobs**
relations β (plain)	1.00 (0.74–1.26)	0.96 (0.65–1.27)
Wiedemann β	2.04 (1.85–2.22)	2.04 (1.85–2.22)
rho GAE β	1.93 (1.77–2.08)	1.93 (1.78–2.09)
LP2 linear algebra β	1.57 (1.12–2.01)	1.70 (1.25–2.15)
crossover $h = 4$ (point; bootstrap; slope band)	$2^{100.4}$; $2^{98.0}$–$2^{103.6}$; $2^{85.9}$–$2^{124.9}$	$2^{97.9}$; $2^{94.6}$–$2^{102.5}$; $2^{82.1}$–$2^{127.4}$
crossover, LP2, $h = 4$	$2^{123.9}$	$2^{127.7}$

5. Predictions

P1 held: 312/312 verified, 290/290 replays reproduced. P2 held ($0.96 \in [0.9, 1.3]$). P3 held (2.04). **P4 not held** (dense elimination 2.34 < 2.4). P5 held (1.93; S_{rho} slope −0.15). P6: crossover $2^{97.9} \in [2^{88}, 2^{115}]$ held; **"no dense crossover below 2^{200}" not held** ($2^{98.9}$, recrossing at 2^{128}). P7 held (1.70; 1.26 > 0.96; every paired ratio at $p \geq 101$ above 1: 1.09–2.31). P8 held.

6. Audit

`ecbench verify` on all 72 sessions (receipts `audits/<job>.audit.json`; scope fixed by amendment 2): **audit OK on all 72, 312 records, 0 problems.** Full replay (`--replay-all`) on 61 sessions, every session with $p \leq 197$, every rho-only session and the spot checks `main-p293-t0` and `main-p401-t0`: **290 replays reproduced exactly**. Integrity only (stored $[d]G = Q$ checks, every planned run recorded, hashes) on the other 11: `main-p1009-t0/t1`, `main-p701-t0/t1`, `main-p509-t0/t1`, `main-p401-t1`, `main-p293-t1/t2`, `lp2-p293-t0`, `lp2-p293-t1`. The first 60 sessions were audited on 2026-10-16 and the last 12 on 2026-10-19, with the same `ecbench` binary.

7. Limits

The exponents are fitted over $p = 29 \ldots 1009$ (groups 2^{19}–2^{40}); the crossover is an extrapolation of those fits, not a measurement, and its width is set by curve-to-curve scatter in the relation rate. The large-prime fits rest on two targets per rung over $p \leq 293$. S is a count on one host class; wall time is not admitted. No `docs/bounds` domain exists for $\mathbb{F}_{p^4}$ curves, so there is no challenge verdict. Study files: `README.md`, `raw/summary.json`, `sessions/`, run reports `reports/run-20261015T0457Z/` and `reports/run-20261019T1612Z/`.